

50-30-20 Budget - Dev

California CCSS Mathematics Standards Alignment

OBJECTIVES	CODE ASSIGNMENTS	HIGH CONFIDENCE	GRADE BAND
27 22 aligned · 5 not	48 23 distinct standards	32 13 med · 3 low	9-12 inferred from context

Auto-curated via run_alignment.py (heuristic mode, no LLM in the loop).

Section 1: Project Setup

Watch Me

Watch the " Watch Me " video linked in cell D1 . This video will introduce you to important GradeFlow features and show you exactly what you need to do to complete this task. You can also watch the task video(s) for step-by-step instructions .

sections[0].objectives[0]

Note: logistical/operational objective, no MATH standard applies

logistical/operational objective, no MATH standard applies

Personal Information

Enter the following information for your budget. Use Gemini (or other AI) and Prompt#001 to research careers and income. In cell C2 , type your first and last name. In cell C3 , type the career you plan to have. In cell C4 , enter your age as a number. In cell C5 , type the city and state (or country) where you plan to live. Watch the task video(s) for step-by-step instructions .

sections[0].objectives[1]

Note: logistical/operational objective, no MATH standard applies

logistical/operational objective, no MATH standard applies

Tax and Income

Use Gemini (or other AI) and Prompt#002 in the Resources File to research effective tax rate and income. In cell C6 , enter your total effective tax rate as a decimal (for example, 0.22 for 22%). In cell C7 , enter your annual gross income (your salary before taxes). Format this cell as currency.

sections[0].objectives[2]

A-SSE.B.3.
C

A

CA-MATH

Grade
9-12

Write expressions in equivalent forms to solve problems.

90 / 100
HIGH

Use the properties of exponents to transform expressions for exponential functions. For example, the expression $1.15t$ can be rewritten as $(1.151/12)12t \approx 1.01212t$ to reveal the approximate equivalent monthly interest rate if the annual rate is 15%. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule: \b(budget|income|tax(es)?|expense|spending|sav(e|

Section 2: Formulas

Annual Net Pay

In cell C8 , write a formula that calculates your estimated annual net pay (income after taxes). Your formula must reference cells C6 and C7 , use multiplication and subtraction operators, and include parentheses for the correct order of operations. Format the cell as currency. EXAMPLE : [Annual Gross Income] * (1-[Effective Tax Rate])

sections[1].objectives[0]

A-SSE.B.3.
C

A

CA-MATH

Grade
9-12

Write expressions in equivalent forms to solve problems.

90 / 100
HIGH

Use the properties of exponents to transform expressions for exponential functions. For example, the expression $1.15t$ can be rewritten as $(1.151/12)12t \approx 1.01212t$ to reveal the approximate equivalent monthly interest rate if the annual rate is 15%. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule: \b(budget|income|tax(es)?|expense|spending|sav(e|

A-SSE.B.4

A

CA-MATH

Grade
9-12

Write expressions in equivalent forms to solve problems.

90 / 100
HIGH

Derive the formula for the sum of a finite geometric series (when the common ratio is not 1), and use the formula to solve problems. For example, calculate mortgage payments. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule: \b(budget|income|tax(es)?|expense|spending|sav(e|

Monthly Net Pay

In cell C9 , write a formula that calculates your estimated monthly net pay by dividing your Estimated Annual Net Pay by 12. Your formula must reference cell C8 and use the division operator. Format the cell as currency. EXAMPLE : [Estimated Annual Net Pay] / [Months in a Year] sections[1].objectives[1]

A-SSE.B.3.
C

A

CA-MATH

Grade
9-12

Write expressions in
equivalent forms to solve
problems.

90 / 100
HIGH

Use the properties of exponents to transform expressions for exponential functions. For example, the expression $1.15t$ can be rewritten as $(1.151/12)12t \approx 1.01212t$ to reveal the approximate equivalent monthly interest rate if the annual rate is 15%. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule:
\\b(budget|income|tax(es)?|expense|spending|sav(e|i

A-SSE.B.4

A

CA-MATH

Grade
9-12

Write expressions in
equivalent forms to solve
problems.

90 / 100
HIGH

Derive the formula for the sum of a finite geometric series (when the common ratio is not 1), and use the formula to solve problems. For example, calculate mortgage payments. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule:
\\b(budget|income|tax(es)?|expense|spending|sav(e|i

Section 3: Needs

Housing

In cell B12 , type either "Rent" or "Mortgage" depending on your housing plan. In cell C12 , enter the monthly dollar amount you expect to pay. Format C12 as currency. Practice writing your own CTF Prompt to help you with the research. Watch the task video for step-by-step instructions .

sections[2].objectives[0]

A-SSE.B.4	A	CA-MATH	Grade 9-12	Write expressions in equivalent forms to solve problems.	90 / 100 HIGH
Derive the formula for the sum of a finite geometric series (when the common ratio is not 1), and use the formula to solve problems. For example, calculate mortgage payments. ■					
WHY THIS MAPS					
grade-band match (grade 9-12); keyword boost applied; matched rule: \b(budget income tax(es)? expense spending sav(e i					

A-SSE.A.1.	A	CA-MATH	Grade 9-12	Interpret the structure of expressions.	90 / 100 HIGH
b	Interpret complicated expressions by viewing one or more of their parts as a single entity. For example, interpret $P(1+r)^n$ as the product of P and a factor not depending on P . ■				
WHY THIS MAPS					
grade-band match (grade 9-12); keyword boost applied; matched rule: \b(budget income tax(es)? expense spending sav(e i					

A-SSE.B.3.	A	CA-MATH	Grade 9-12	Write expressions in equivalent forms to solve problems.	90 / 100 HIGH
c	Use the properties of exponents to transform expressions for exponential functions. For example, the expression $1.15t$ can be rewritten as $(1.151/12)^{12t} \approx 1.012^{12t}$ to reveal the approximate equivalent monthly interest rate if the annual rate is 15%. ■				
WHY THIS MAPS					
grade-band match (grade 9-12); keyword boost applied; matched rule: \b(budget income tax(es)? expense spending sav(e i					

Needs List

List at least SEVEN needs in cells B12 through B26 . Include notes to help your teacher and AI understand the particulars of the cost. For example, a few words describing your housing, vehicle, etc... Write a C/T/F Prompt (context, task, format) to help you research 'Need' expenses and their expected amounts based on your location, salary, and personal preferences. For each 'Need', enter the estimated monthly cost in the matching row of column C . Format all dollar amounts as currency. Research realistic costs for your chosen location. Use actual prices or reasonable estimates based on the local cost of living, not guesses. Your listed needs and their monthly costs will be reviewed using AI to check whether they are realistic for your career, salary, and location.

sections[2].objectives[1]

A-CED.A.3	A	CA-MATH	Grade 9-12	Create equations that describe numbers or relationships.	90 / 100 HIGH
Represent constraints by equations or inequalities, and by systems of equations and/or inequalities, and interpret solutions as viable or nonviable options in a modeling context. For example, represent inequalities describing nutritional and cost constraints on combinations of different foods. ■					
WHY THIS MAPS grade-band match (grade 9-12); keyword boost applied; matched rule: \b(budget income tax(es)? expense spending sav(e i					
A-SSE.B.3.	A	CA-MATH	Grade 9-12	Write expressions in equivalent forms to solve problems.	90 / 100 HIGH
C					
Use the properties of exponents to transform expressions for exponential functions. For example, the expression $1.15t$ can be rewritten as $(1.151/12)^{12t} \approx 1.012^{12t}$ to reveal the approximate equivalent monthly interest rate if the annual rate is 15%. ■					
WHY THIS MAPS grade-band match (grade 9-12); keyword boost applied; matched rule: \b(budget income tax(es)? expense spending sav(e i					

Sum Function

In cell D27 , use the SUM() function to add up all of your monthly need costs. The value1 argument must be the range of C12:C26 . Format the cell as currency. EXAMPLE : SUM (value1)

sections[2].objectives[2]

F-IF.A.1	F	CA-MATH	Grade 9-12	Understand the concept of a function and use function notation.	90 / 100 HIGH
Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range. If f is a function and x is an element of its domain, then $f(x)$ denotes the output of f corresponding to the input x. The graph of f is the graph of the equation $y = f(x)$.					
WHY THIS MAPS grade-band match (grade 9-12); keyword boost applied; matched rule: $\backslash b(\text{add}(\text{ing} \text{ition})?\text{subtract}(\text{ing} \text{ion})?\text{multipl}(\text{y} \text{ic}$					
F-IF.A.2	F	CA-MATH	Grade 9-12	Understand the concept of a function and use function notation.	90 / 100 HIGH
Use function notation, evaluate functions for inputs in their domains, and interpret statements that use function notation in terms of a context.					
WHY THIS MAPS grade-band match (grade 9-12); keyword boost applied; matched rule: $\backslash b(\text{add}(\text{ing} \text{ition})?\text{subtract}(\text{ing} \text{ion})?\text{multipl}(\text{y} \text{ic}$					
F-BF.A.1	F	CA-MATH	Grade 9-12	Build a function that models a relationship between two quantities.	90 / 100 HIGH
Write a function that describes a relationship between two quantities. ■					
WHY THIS MAPS grade-band match (grade 9-12); keyword boost applied; matched rule: $\backslash b(\text{add}(\text{ing} \text{ition})?\text{subtract}(\text{ing} \text{ion})?\text{multipl}(\text{y} \text{ic}$					

% Monthly Net Pay

In cell E27 , write a formula that divides your Need Subtotal (D27) by your Estimated Monthly Net Pay (C9). Use an absolute row reference for Estimated Monthly Net Pay so this formula still works if you copy it to other subtotal rows later. Format the cell as a percentage. EXAMPLE : [Need Subtotal] / [Estimated Monthly Net Pay]

sections[2].objectives[3]

A-SSE.B.3.C

A

CA-MATH

Grade 9-12

Write expressions in equivalent forms to solve problems.

90 / 100
HIGH

Use the properties of exponents to transform expressions for exponential functions. For example, the expression $1.15t$ can be rewritten as $(1.151/12)12t \approx 1.01212t$ to reveal the approximate equivalent monthly interest rate if the annual rate is 15%. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule: $\backslash b(ratio|proportion(al)?|percent(age)?|unit rate|p$

A-SSE.B.4

A

CA-MATH

Grade 9-12

Write expressions in equivalent forms to solve problems.

90 / 100
HIGH

Derive the formula for the sum of a finite geometric series (when the common ratio is not 1), and use the formula to solve problems. For example, calculate mortgage payments. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule: $\backslash b(ratio|proportion(al)?|percent(age)?|unit rate|p$

Section 4: Wants

Entertainment

In cell B30 , type "Entertainment." In cell C30 , enter the monthly dollar amount you plan to spend on entertainment. Format C30 as currency.

sections[3].objectives[0]

A-SSE.B.3.C

A

CA-MATH

Grade 9-12

Write expressions in equivalent forms to solve problems.

70 / 100
MEDIUM

Use the properties of exponents to transform expressions for exponential functions. For example, the expression $1.15t$ can be rewritten as $(1.151/12)12t \approx 1.01212t$ to reveal the approximate equivalent monthly interest rate if the annual rate is 15%. ■

WHY THIS MAPS

grade-band match (grade 9-12)

Wants List

List at least SEVEN wants in cells B30 through B44 . For each want, enter the estimated monthly cost in the matching row of column C . Format all dollar amounts as currency. Include notes to help your teacher and AI understand the particulars of the cost. For example, a few words describing what your entertainment, travel, shopping, etc.... looks like.

sections[3].objectives[1]

A-CED.A.3	A	CA-MATH	Grade 9-12	Create equations that describe numbers or relationships.	70 / 100 MEDIUM
Represent constraints by equations or inequalities, and by systems of equations and/or inequalities, and interpret solutions as viable or nonviable options in a modeling context. For example, represent inequalities describing nutritional and cost constraints on combinations of different foods. ■					
WHY THIS MAPS					
grade-band match (grade 9-12)					

G-MG.A.3	G	CA-MATH	Grade 9-12	Apply geometric concepts in modeling situations.	70 / 100 MEDIUM
Apply geometric methods to solve design problems (e.g., designing an object or structure to satisfy physical constraints or minimize cost; working with typographic grid systems based on ratios). ■					
WHY THIS MAPS					
grade-band match (grade 9-12)					

A-SSE.B.3. C	A	CA-MATH	Grade 9-12	Write expressions in equivalent forms to solve problems.	50 / 100 LOW
Use the properties of exponents to transform expressions for exponential functions. For example, the expression $1.15t$ can be rewritten as $(1.151/12)^{12t} \approx 1.012^{12t}$ to reveal the approximate equivalent monthly interest rate if the annual rate is 15%. ■					
WHY THIS MAPS					
grade-band match (grade 9-12)					

Reusing Formulas

In this task you'll learn how to copy and paste cell formulas and formatting for use in other areas or your spreadsheet. Watch the task video(s) for step-by-step instructions and to learn why this method works in this situation . Select cells D27 and E27 , and copy them. Select cells D45 and E45 , and paste. If you want be build the formula manually, follow these instructions. In cell D45 , use the SUM() function to add up all of your monthly want costs. Your formula should sum the range C30:C44 . Format the cell as currency. EXAMPLE : SUM (value1) sections[3].objectives[2]

% Monthly Net Pay

In this task you'll learn how to copy and paste cell formulas and formatting for use in other areas or your spreadsheet. Watch the task video(s) for step-by-step instructions and to learn why this method works in this situation . Select cells D27 and E27 , and copy them. Select cells D45 and E45 , and paste. If you want be build the formula manually, follow these instructions. In cell E45 , write a formula that divides your Wants Subtotal (D45) by your Estimated Monthly Net Pay. Use the absolute row reference C\$9 . Format the cell as a percentage. EXAMPLE : [Want Subtotal] / [Estimated Monthly Net Pay] sections[3].objectives[3]

F-BF.A.2

F

CA-MATH

Grade 9-12

Build a function that models a relationship between two quantities.

90 / 100 HIGH

Write arithmetic and geometric sequences both recursively and with an explicit formula, use them to model situations, and translate between the two forms. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule: \b(add(ing|ition)?|subtract(ing|ion)?|multipl(y|ic

F-IF.A.1

F

CA-MATH

Grade 9-12

Understand the concept of a function and use function notation.

90 / 100 HIGH

Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range. If f is a function and x is an element of its domain, then f(x) denotes the output of f corresponding to the input x. The graph of f is the graph of the equation y = f(x).

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule: \b(add(ing|ition)?|subtract(ing|ion)?|multipl(y|ic

A-SSE.B.3. C

A

CA-MATH

Grade 9-12

Write expressions in equivalent forms to solve problems.

90 / 100 HIGH

Use the properties of exponents to transform expressions for exponential functions. For example, the expression 1.15t can be rewritten as (1.151/12)12t ≈ 1.01212t to reveal the approximate equivalent monthly interest rate if the annual rate is 15%. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule: \b(ratio|proportion(al)?|percent(ge)?|unit rate|p

A-SSE.B.4

A

CA-MATH

Grade 9-12

Write expressions in equivalent forms to solve problems.

90 / 100 HIGH

Derive the formula for the sum of a finite geometric series (when the common ratio is not 1), and use the formula to solve problems. For example, calculate mortgage payments. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule: \b(ratio|proportion(al)?|percent(ge)?|unit rate|p

Section 5: Financials

Emergency Fund

In cell B48 , type "Emergency Fund."
In cell C48 , enter the monthly dollar amount you plan to set aside for emergencies. Format C48 as currency.
sections[4].objectives[0]

A-SSE.B.3.C	A	CA-MATH	Grade 9-12	Write expressions in equivalent forms to solve problems.	70 / 100 MEDIUM
Use the properties of exponents to transform expressions for exponential functions. For example, the expression $1.15t$ can be rewritten as $(1.151/12)12t \approx 1.01212t$ to reveal the approximate equivalent monthly interest rate if the annual rate is 15%. ■					
WHY THIS MAPS					
grade-band match (grade 9-12)					

F-IF.A.1	F	CA-MATH	Grade 9-12	Understand the concept of a function and use function notation.	70 / 100 MEDIUM
Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range. If f is a function and x is an element of its domain, then $f(x)$ denotes the output of f corresponding to the input x . The graph of f is the graph of the equation $y = f(x)$.					
WHY THIS MAPS					
grade-band match (grade 9-12)					

A-REI.D.10	A	CA-MATH	Grade 9-12	Represent and solve equations and inequalities graphically.	70 / 100 MEDIUM
Understand that the graph of an equation in two variables is the set of all its solutions plotted in the coordinate plane, often forming a curve (which could be a line).					
WHY THIS MAPS					
grade-band match (grade 9-12)					

Financials List

List at least FOUR financial goals in cells B48 through B52 . For each goal, enter the estimated monthly contribution in the matching row of column C . Format all dollar amounts as currency. Include notes to help your teacher and AI understand the particulars of the cost. For example, a few words describing what your emergency fund, retirement accounts, and financial goals look like. Write a CTF prompt to help or use the prompt used for the Wants and ask AI to rewrite it for use on Financial Goals section.

sections[4].objectives[1]

A-SSE.B.3.C	A	CA-MATH	Grade 9-12	Write expressions in equivalent forms to solve problems.	70 / 100 MEDIUM
Use the properties of exponents to transform expressions for exponential functions. For example, the expression $1.15t$ can be rewritten as $(1.151/12)12t \approx 1.01212t$ to reveal the approximate equivalent monthly interest rate if the annual rate is 15%. ■					
WHY THIS MAPS					
grade-band match (grade 9-12)					

A-CED.A.3	A	CA-MATH	Grade 9-12	Create equations that describe numbers or relationships.	70 / 100 MEDIUM
Represent constraints by equations or inequalities, and by systems of equations and/or inequalities, and interpret solutions as viable or nonviable options in a modeling context. For example, represent inequalities describing nutritional and cost constraints on combinations of different foods. ■					
WHY THIS MAPS					
grade-band match (grade 9-12)					

N-CN.C.8	N	CA-MATH	Grade 9-12	Use complex numbers in polynomial identities and equations.	50 / 100 LOW
Extend polynomial identities to the complex numbers. For example, rewrite $x^2 + 4$ as $(x + 2i)(x - 2i)$.					
WHY THIS MAPS					
grade-band match (grade 9-12)					

Modify Copied Formulas

In this task you'll learn how to copy, paste, and adjust cell formulas and formatting in other areas or your spreadsheet. Watch the task video(s) for step-by-step instructions and to learn why this method works in this situation . Select cells D45 and E45 , and copy them. Select cells D53 and E53 , and paste. If you want to build the formula manually, follow these instructions. In cell D53 , use the SUM() function to add up all of your monthly financial goal contributions. Your formula should sum the range C48:C52 . Format the cell as currency. EXAMPLE : SUM (value1) sections[4].objectives[2]

F-BF.A.2

F

CA-MATH

Grade
9-12

Build a function that models a relationship between two quantities.

90 / 100
HIGH

Write arithmetic and geometric sequences both recursively and with an explicit formula, use them to model situations, and translate between the two forms. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule:
\\b(add(ing|ition)?|subtract(ing|ion)?|multipl(y|ic

F-IF.A.1

F

CA-MATH

Grade
9-12

Understand the concept of a function and use function notation.

90 / 100
HIGH

Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range. If f is a function and x is an element of its domain, then $f(x)$ denotes the output of f corresponding to the input x . The graph of f is the graph of the equation $y = f(x)$.

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule:
\\b(add(ing|ition)?|subtract(ing|ion)?|multipl(y|ic

F-BF.A.1.b

F

CA-MATH

Grade
9-12

Build a function that models a relationship between two quantities.

90 / 100
HIGH

Combine standard function types using arithmetic operations. For example, build a function that models the temperature of a cooling body by adding a constant function to a decaying exponential, and relate these functions to the model. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule:
\\b(add(ing|ition)?|subtract(ing|ion)?|multipl(y|ic

% Monthly Net Pay

In this task you'll learn how to copy, paste, and adjust cell formulas and formatting in other areas or your spreadsheet. Watch the task video(s) for step-by-step instructions and to learn why this method works in this situation . Select cells D45 and E45 , and copy them. Select cells D53 and E53 , and paste. If you want be build the formula manually, follow these instructions. In cell E53 , write a formula that divides your Financial Goals Subtotal (D53) by your Estimated Monthly Net Pay. Use the absolute row reference C\$9 . Format the cell as a percentage. EXAMPLE : [Financials Subtotal] / [Estimated Monthly Net Pay]

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sections[4].objectives[3]
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A-SSE.B.3.C

A

CA-MATH

Grade
9-12

Write expressions in equivalent forms to solve problems.

90 / 100
HIGH

Use the properties of exponents to transform expressions for exponential functions. For example, the expression $1.15t$ can be rewritten as $(1.151/12)12t \approx 1.01212t$ to reveal the approximate equivalent monthly interest rate if the annual rate is 15%. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule:
 $\backslash b(\text{ratio}|\text{proportion}(\text{al})?\text{percent}(\text{age})?\text{unit rate}|p$

A-SSE.B.4

A

CA-MATH

Grade
9-12

Write expressions in equivalent forms to solve problems.

90 / 100
HIGH

Derive the formula for the sum of a finite geometric series (when the common ratio is not 1), and use the formula to solve problems. For example, calculate mortgage payments. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule:
 $\backslash b(\text{ratio}|\text{proportion}(\text{al})?\text{percent}(\text{age})?\text{unit rate}|p$

Section 6: Totals and Chart

Total Spending

In cell D55 , write a formula that adds your three subtotals together: Needs Subtotal (D27), Wants Subtotal (D45), and Financial Goals Subtotal (D53). Use the addition operator. Format the cell as currency. EXAMPLE : [Needs Subtotal] + [Wants Subtotal] + [Financials Subtotal]

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sections[5].objectives[0]
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A-SSE.B.4

A

CA-MATH

Grade
9-12

Write expressions in equivalent forms to solve problems.

90 / 100
HIGH

Derive the formula for the sum of a finite geometric series (when the common ratio is not 1), and use the formula to solve problems. For example, calculate mortgage payments. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule:
 $\backslash b(\text{budget}|\text{income}|\text{tax}(\text{es})?\text{expense}|\text{spending}|\text{sav}(\text{e})|i$

Remaining Funds

In cell D56 , write a formula that subtracts your Total Spending (D55) from your Estimated Monthly Net Pay (C9). This shows how much money is left over (or how much you are over budget). Format the cell as currency. In cells E55 and E56 , write formulas that divide each row's dollar amount by your Estimated Monthly Net Pay using the absolute row reference C\$9 . Format both cells as percentages. EXAMPLE : [Total Spending] / [Estimated Monthly Net Pay] [Remaining Funds] / [Estimated Monthly Net Pay] sections[5].objectives[1]

A-SSE.B.3.
C

A

CA-MATH

Grade
9-12

Write expressions in equivalent forms to solve problems.

90 / 100
HIGH

Use the properties of exponents to transform expressions for exponential functions. For example, the expression $1.15t$ can be rewritten as $(1.151/12)12t \approx 1.01212t$ to reveal the approximate equivalent monthly interest rate if the annual rate is 15%. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule:
`\b(budget|income|tax(es)?|expense|spending|sav(e|`

A-SSE.B.4

A

CA-MATH

Grade
9-12

Write expressions in equivalent forms to solve problems.

90 / 100
HIGH

Derive the formula for the sum of a finite geometric series (when the common ratio is not 1), and use the formula to solve problems. For example, calculate mortgage payments. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule:
`\b(budget|income|tax(es)?|expense|spending|sav(e|`

S-IC.A.2

S

CA-MATH

Grade
9-12

Understand and evaluate random processes underlying statistical experiments.

90 / 100
HIGH

Decide if a specified model is consistent with results from a given data-generating process, e.g., using simulation. For example, a model says a spinning coin falls heads up with probability 0.5. Would a result of 5 tails in a row cause you to question the model? ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule:
`\b(budget|income|tax(es)?|expense|spending|sav(e|`

Pie Chart

Create a pie chart that visualizes how your budget is divided across Needs, Wants, Financial Goals, and Remaining Funds. Select your four sets of cells (label and value), then and insert a chart. Setup: In the Chart Editor, choose any pie chart style. Set "Combine ranges" to "Vertically" so all four categories appear as slices in a single chart. Customize: Change the chart title to 50/30/20: [Your Name] (use your actual name). Move the chart legend to either the right or left side. Under the Pie Chart settings, turn on labels and set them to show percentages. Watch the task video for step-by-step instructions.

sections[5].objectives[2]

S-ID.A.4

S

CA-MATH

Grade 9-12

Summarize, represent, and interpret data on a single count or measurement variable.

90 / 100
HIGH

Use the mean and standard deviation of a data set to fit it to a normal distribution and to estimate population percentages. Recognize that there are data sets for which such a procedure is not appropriate. Use calculators, spreadsheets, and tables to estimate areas under the normal curve. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule: \b(budget|income|tax(es)?|expense|spending|sav(e|i

A-CED.A.2

A

CA-MATH

Grade 9-12

Create equations that describe numbers or relationships.

90 / 100
HIGH

Create equations in two or more variables to represent relationships between quantities; graph equations on coordinate axes with labels and scales. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule: \b(budget|income|tax(es)?|expense|spending|sav(e|i

Section 7: Reflections and Notes

Notes

You must complete the "Notes" section for at least EIGHT items in your budget. Focus on the ones that are the highest dollar amounts or most general. For example, housing, entertainment, vehicle, etc.... Include notes that help your teacher and an AI to understand the dollar amount. For example; the year, make, and model of a car. The number of rooms, SQFT, and if you have roommates for housing. Also what you will do with your entertainment or travel funds.

sections[6].objectives[0]

MP.4

MP

CA-MATH

Grade — Mathematical Practice

70 / 100
MEDIUM

Model with mathematics.

WHY THIS MAPS

grade-band match (grade None); keyword boost applied; matched rule: \b(budget|income|tax(es)?|expense|spending|sav(e|

MP.7

MP

CA-MATH

Grade — Mathematical Practice

70 / 100
MEDIUM

Look for and make use of structure.

WHY THIS MAPS

grade-band match (grade None); keyword boost applied; matched rule: \b(budget|income|tax(es)?|expense|spending|sav(e|

G-GPE.A.2

G

CA-MATH

Grade 9-12

Translate between the geometric description and the equation for a conic section.

70 / 100
MEDIUM

Derive the equation of a parabola given a focus and directrix.

WHY THIS MAPS

grade-band match (grade 9-12); matched rule: \b(budget|income|tax(es)?|expense|spending|sav(e|

Spreadsheet Reflection

Write a reflection (at least 25 words) about what you learned from building this spreadsheet. What spreadsheet skills were new to you? What did you find most helpful? What did you find most difficult?

sections[6].objectives[1]

Note: reflective/metacognitive objective, no MATH standard applies

reflective/metacognitive objective, no MATH standard applies

50/30/20 Reflection

Write a reflection (at least 25 words) about what you learned from building a 50/30/20 budget based on income and research. What did you find most interesting? What did you find most helpful? What did you find most difficult?

sections[6].objectives[2]

Note: reflective/metacognitive objective, no MATH standard applies

reflective/metacognitive objective, no MATH standard applies

AI Reflection

Write a reflection (at least 25 words) about what you learned from building a 50/30/20 budget based on income and research. What did you find most interesting? What did you find most helpful? What did you find most difficult?

sections[6].objectives[3]

Note: reflective/metacognitive objective, no MATH standard applies

reflective/metacognitive objective, no MATH standard applies

Section 8: Teacher Graded

Budget Realism

Your budget should tell a believable financial story. The career you chose, the location you picked, the salary you researched, and the expenses you listed should all connect logically. A realistic budget is not about being perfect. It is about showing that you thought carefully about what life actually costs for someone in your situation.

sections[7].objectives[0]

F-LE.B.5

F

CA-MATH

Grade 9-12

Interpret expressions for functions in terms of the situation they model.

90 / 100
HIGH

Interpret the parameters in a linear or exponential function in terms of a context. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule: \b(budget|income|tax(es)?|expense|spending|sav(e)|

Reflection Quality

Your three reflections are a chance to show what you actually took away from this project. Strong reflections describe specific things you learned, challenges you faced, or decisions you made. They go beyond general statements and show that you are thinking critically about your own learning.

sections[7].objectives[1]

S-MD.A.2

S

CA-MATH

Grade 9-12

Calculate expected values and use them to solve problems.

90 / 100
HIGH

Calculate the expected value of a random variable; interpret it as the mean of the probability distribution. ■

WHY THIS MAPS

grade-band match (grade 9-12); keyword boost applied; matched rule: \b(probabilit(y|ies)|likelihood|chance|random|simu

Accuracy and Completeness

The 50/30/20 rule only works if items are placed in the right category. This task checks whether your needs, wants, and financial goals are properly sorted and whether your overall budget shows an intentional plan rather than random numbers. The pie chart configuration is included here as well.

sections[7].objectives[2]

G-SRT.8.1	G	CA-MATH	Grade 9-12	Define trigonometric ratios and solve problems involving right triangles.	70 / 100 MEDIUM
Derive and use the trigonometric ratios for special right triangles (30°, 60°, 90° and 45°, 45°, 90°)					
WHY THIS MAPS					
grade-band match (grade 9-12); matched rule: \\b(budget income tax(es)? expense spending sav(e					

N-Q.A.3	N	CA-MATH	Grade 9-12	Reason quantitatively and use units to solve problems.	70 / 100 MEDIUM
Choose a level of accuracy appropriate to limitations on measurement when reporting quantities. ■					
WHY THIS MAPS					
grade-band match (grade 9-12); matched rule: \\b(budget income tax(es)? expense spending sav(e					

7.RP.A.2.c	RP	CA-MATH	Grade 7	Analyze proportional relationships and use them to solve real-world and mathematical problems.	50 / 100 LOW
Represent proportional relationships by equations. For example, if total cost t is proportional to the number n of items purchased at a constant price p, the relationship between the total cost and the number of items can be expressed as t = pn.					
WHY THIS MAPS					
keyword boost applied; matched rule: \\b(budget income tax(es)? expense spending sav(e					